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* Reinforcement learning uses a feedback / reward system.
* Feedback can be ⊕ or ⊖
  - Positive reward - desirable action
  - Negative reward / penalty - undesirable action.

Boston Dynamics - robots. (Robotics company) - First they made robots for millitory. 4 leg robots like dogs, etc.

* Robots can use reinforcement learning concepts to learn suitable actions through interaction and feedback.

Real coded genetic algorithm.

This represents solutions using real valued numbers rather than only binary values.

ex: (0.25, 1.73, 4.62, 2.15)

This is useful when the decision variables are naturally continuous.

* Supervised ML is mostly predictive.
* Unsupervised - Generative.
* Generative ML = Unsupervised ML + Supervised fine tuning + Reinforcement learning

Supervised ML

- → Predicting a class (Classification)
- → Predicting a number (Regression)
- → Predicting a value from a time series.

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Reinforcement Learning (RL)

Reinforcement Learning is a type of ML where an agent learns by interacting with an environment and receiving feedback in the form of rewards (+) or penalties (−).

ex: • Robots developed by Boston Dynamics learn movements and actions through interaction & feedback.

RL representation

State (s) → Current situation

Action (a) → Action taken by the agent.

Policy (π) → Rule used to select actions.

Probability (P) → Probability of selecting the action.

$$\pi(s_1, a_1) = P_1$$

Unsupervised Machine Learning

Works without target lables.

1 - Clustering

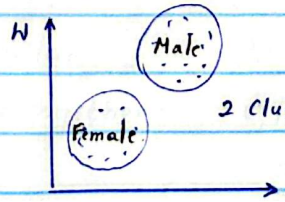
2 - Association Rule Mining.

Clustering

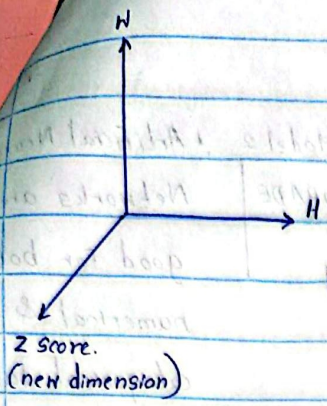
Clustering is the process of grouping similar data items together without using predefined lables.

Student data

Height	Weight	Z score	GPA 1	GPA 2	GPA 3



If we take only these two dimensions it will create two clusters. We can predict those two are male & female.



With this new dimension existing clusters will disappear & it will make new clusters.

If there are many more dimensions → High dimensional space.

Curse of dimensionality → When the number of dimensions increases :

- Data becomes sparse.
- Distance calculations become difficult.
- Learning becomes more complex.

Association Rule Mining

This discovers relationships among data items.

Students who completed degree.

				Final GPA	Grade
				0-4	IC/2L/2U/S/F

← Predicting a category (Classification)

↑ Predicting a numeric value. (Regression)

x_1	x_2	x_3	GPA	Grade
	Training data			
	Test	data		

80 X. ← (S) ← Population.

20 X.

[M] This ML model trained by using sample data.

If one sample trained too much it may perform poorly on other samples. This is the bias-variance problem. (Model overfitted to one sample).

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For prediction we need techniques.

→ Linear Regression
→ Logistic Regression

Model 1	Model 2
GPA	GRADE

* Artificial Neural Networks are good for both numerical & categorical data.

- Linear Regression → Simple (one input feature)
  → Multiple. (Multiple input features)

$X = \{x_1, x_2, x_3\}$ $\quad y_1 \propto X$

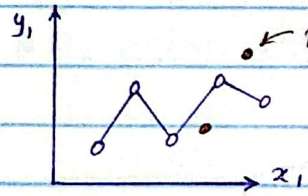
We need to find the relationship between x and y

$y = f(x)$

- Feature engineering
- Feature selecting

} Data preprocessing methods.

For one feature: $X = \{x_1\}$

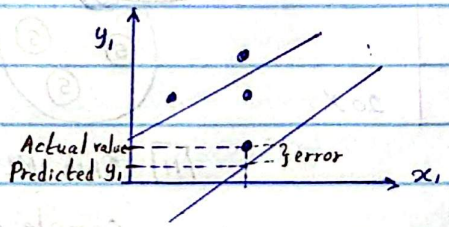


* Model overfitted to train data.

$\hat{y} = W_1 x_1 + W_0$

W_1 - Weight

W_0 - bias.



By changing W_1 and W_0, we can draw an infinite number of possible lines. We need to find the best one.

Error at each point - difference between the actual value and the predicted value.

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Error function (Sum of Squared Error - SSE)

$$E = \varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 + \dots = SSE$$

$$E = \sum_{i=1}^{n} (y - \hat{y}^{(i)})^2$$

Since $\hat{y} = w_1 x_1 + w_0$, this becomes

$$E = \sum_{i=1}^{n} (y - (w_1 x_1 + w_0))^2$$

$$E = f(w_1, w_0)$$

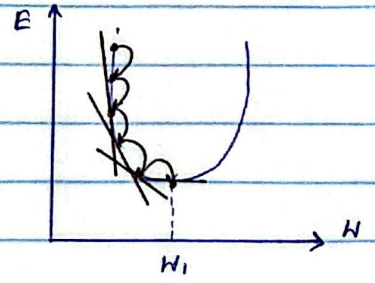
* Different values of (w_0, w_1) give different amounts of error.
* ML = finding the parameter values $(w_0, w_1, \dots)$ that minimize E.
* That is the line with the least total squared error is the optimal line.

2 features: $X = \{x_1, x_2\}$

$$\hat{y} = w_1 x_1 + w_2 x_2 + w_0$$

$$E = \sum_{i=1}^{n} (y - (w_1 x_1 + w_2 x_2 + w_0))^2$$

$$E = f(w_0, w_1, w_2)$$



* Plotting error (E) vs. Height (W) typically produces a convex curve.
* The algorithm takes small steps (jumps) down the curve toward the minimum error.
* Large jumps - faster but risk overshooting the minimum.
* Small jumps - more precise but takes much longer to converge.

η = Learning rate - controls the step size taken at each iteration while minimizing E.

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